

REFERENSI

- Acharya, U. R., Fujita, H., Lih, O. H., Adam, M., Tan, J. H., & Chua, C. K. (2017). *Automated Arrhythmia detection using spectrogram and deep convolutional neural network with long duration ECG signals. Information Sciences*, 405, 112–127.
- Acharya, U. R., Joseph, K. P., Kannathal, N., Lim, C. M., & Suri, J. S. (2006). *Heart rate variability: A review. Medical & Biological Engineering & Computing*, 44(12), 1031–1051.
- Acharya, U. R., Oh, S. L., Hagiwara, Y., Tan, J. H., & Adam, M. (2017). *A deep convolutional neural network model to classify heartbeats. Computers in Biology and Medicine*, 89, 389–396.
- Ahmed, M. S., Khan, A. S., & Ali, N. (2023). *ECG signal classification using deep learning techniques: A systematic review. Computers in Biology and Medicine*, 161, 107380.
- Ahmed, M. S., Khan, A. S., & Ali, N. (2025). *Detection of normal and abnormal ECG signals using a 1D convolutional neural network on a modified MIT-BIH dataset (Unpublished master's thesis). Universitas Bina Nusantara, Bandung.*
- Alinsaif, S. (2024). *Unraveling arrhythmias with graph-based analysis: A survey of the MIT-BIH database. Computation*, 12(2), 21. <https://doi.org/10.3390/computation12020021>.
- American Heart Association. (2021). *What is Arrhythmia? Retrieved from https://www.heart.org/en/health-topics/Arrhythmia/about-Arrhythmia*
- Bayat, O., Aljawarneh, S., Carlak, H. F., International Association of Researchers, Institute of Electrical and Electronics Engineers, & Akdeniz Üniversitesi. (2017). *Understanding of a convolutional neural network. In IEEE International Conference on Engineering & Technology (ICET) (pp. 21–23). Antalya.*
- Chen, Y. (2015). *Convolutional neural network for sentence classification. Ontario.*
- Chicco, D., & Jurman, G. (2020). *The advantages of the Matthews correlation coefficient over F1 score and accuracy in binary classification evaluation. BMC Genomics*, 21(1), 6.
- Clifford, G. D., Azuaje, F., & McSharpy, P. (2006). *Advanced methods and tools for ECG data analysis. Artech House.*
- Eleyan, A., & Alboghbaish, E. (2024). *Electrocardiogram signals classification using deep-learning-based incorporated convolutional neural network and long short-term memory framework. IEEE Access*, 12, 14223–14232. <https://doi.org/10.1109/ACCESS.2024.3334562>.

- Elgendi, M., Jonkman, M., & De Boer, F. (2014). Frequency Bands Effects on QRS Detection. *Biosensors using Bio-electronics, Control, and Signal Processing*, 428-431.
- Faust, O., Hagiwara, Y., Hong, T. J., Lih, O. S., & Acharya, U. R. (2018). *Deep learning for healthcare applications based on physiological signals: A review. Computer Methods and Programs in Biomedicine*, 161, 1–13.
- Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., ... & Stanley, H. E. (2000). *PhysioBank, PhysioToolkit, and PhysioNet. Circulation*, 101(23), e215–e220.
- Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., Mietus, J. E., Moody, G. B., Peng, C. K., & Stanley, H. E. (2000). *PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation*, 101(23), e215–e220. <https://doi.org/10.1161/01.CIR.101.23.e215>
- Guerra, I., Castro, J. M., Silva, R. A., & Fonseca, J. M. (2025). *Deep learning for ECG signal analysis: A systematic review. Biomedical Signal Processing and Control*, 97, 106367.
- Hannun, A. Y., Rajpurkar, P., Haghpanahi, M., Tison, G. H., Bourn, C., Turakhia, M. P., & Ng, A. Y. (2019). *Cardiologist-level Arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. Nature Medicine*, 25(1), 65–69.
- Hinton, G. E., & Salakhutdinov, R. R. (2006). *Reducing the dimensionality of data with neural networks. Science*, 313(5786), 504–507.
- Jain, A. K. (2010). Data clustering: 50 years beyond K-means. *Pattern Recognition Letters*, 31(8), 651–666.
- Kwon, S., Hong, J., & Park, Y. (2018). *Smart wearable systems for personalized health monitoring. IEEE Reviews in Biomedical Engineering*, 11, 356–367.
- Lloyd, S. (1982). Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2), 129–137.
- Luz, E. J. d. S., Schwartz, W. R., Cámara-Chávez, G., & Menotti, D. (2016). *ECG-based heartbeat classification for Arrhythmia detection: A survey. Computer Methods and Programs in Biomedicine*, 127, 144–164.
- Martis, R. J., Acharya, U. R., Adeli, H., & Prasad, H. (2014). *Application of higher order statistics for atrial Arrhythmia classification. Biomedical Signal Processing and Control*, 8(6), 888–900.
- Moody, G. B., & Mark, R. G. (2001). *The impact of the MIT-BIH Arrhythmia Database. IEEE Engineering in Medicine and Biology Magazine*, 20(3), 45–50.

- Nab, A. (2023). *ECG Arrhythmia classification using deep learning (Unpublished master's thesis)*. University of Twente.
- Nehru Institute of Technology, P. J., & R. K. (2025). *Early detection of cardiac Arrhythmia using deep learning. Journal of Medical Engineering & Technology*, 49(2), 118–124.
- Pan, J., & Tompkins, W. J. (1985). A real-time QRS detection algorithm. *IEEE Transactions on Biomedical Engineering*, (3), 230-236.
- Panwar, M., Singh, S., & Singh, R. (2025). *Real-time ECG Arrhythmia detection using deep learning. Biomedical Signal Processing and Control*, 95, 106300.
- Pantelopoulos, A., & Bourbakis, N. G. (2010). *A survey on wearable sensor-based systems for health monitoring and prognosis. IEEE Transactions on Systems, Man, and Cybernetics*, 40(1), 1–12.
- Rajpurkar, P., Hannun, A. Y., Haghpanahi, M., Bourn, C., & Ng, A. Y. (2017). *Cardiologist-level Arrhythmia detection with convolutional neural networks. Nature Medicine*, 25(1), 65–69.
- Sannino, G., & De Pietro, G. (2021). *Deep learning for ECG signal classification: A review. Artificial Intelligence in Medicine*, 118, 102142.
- Shaffer, F., & Ginsberg, J. P. (2017). *An overview of heart rate variability metrics and norms. Frontiers in Public Health*, 5, 258. <https://doi.org/10.3389/fpubh.2017.00258>
- Shi, H., Zhang, C., He, J., & Wang, Z. (2019). *Automatic detection of Arrhythmia based on multi-scale CNN and attention-based RNN. Journal of Biomedical Informatics*, 100, 103395.
- Taejoong Yoon. (2023). *MIT-BIH Arrhythmia Dataset. Kaggle*. <https://www.kaggle.com/datasets/taejoongyoon/mitbit-Arrhythmia-database>
- Xiong, Z., Stiles, M. K., & Zhao, J. (2021). *A deep learning framework for automatic diagnosis of Arrhythmias using a single-lead ECG. Journal of Electrocardiology*, 66, 29–35.
- Xu, R., & Wunsch, D. (2005). Survey of clustering algorithms. *IEEE Transactions on Neural Networks*, 16(3), 645–678.
- Yildirim, O., Baloglu, U. B., Tan, R. S., & Acharya, U. R. (2018). *Arrhythmia detection using deep convolutional neural network with long duration ECG signals. Computers in Biology and Medicine*, 102, 411–420.
- Zhao, Z., Zhang, Y., Deng, Y., & Zhou, X. (2019). *ECG signal denoising and classification using deep feature learning. Medical & Biological Engineering & Computing*, 57, 1987–1998.

- Zheng, J., Zhang, Y., Wang, L., Chen, H., & Wu, X. (2020). *A deep learning framework for ECG-based heartbeat classification using spatial pyramid pooling*. *Neural Networks*, 122, 160–169.
- Zipes, D. P., & Jalife, J. (2013). *Cardiac electrophysiology: From cell to bedside (6th ed.)*. Elsevier.